

A PROJECT REPORT
ON
“FOOD MANAGEMENT”

**Submitted in partial fulfillment of the requirements for the award Bachelor of
Computer Application of KANNUR UNIVERSITY *BY***

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Under the guidance of

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AMSTECK ARTS AND SCIENCE COLLEGE KALLIASSERI

(Affiliated to Kannur University)

Bachelor of Computer Application

(2018-2021)

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(2018-2021)



CERTIFICATE

This is to certify that the project entitled “FOOD MANAGEMENT” is to bonafide record of the work done by ADITH KUMAR T

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DECLARATION

I hereby declare that the project title entitled "FOOD MANAGEMENT" submitted by ADITH KUMAR T bearing Register Number AM18BCAR07 to Kannur University in Partial fulfillment of the requirement for the award of the Bachelor of Computer Application is Record of bonafide project work carried out by me under the guidance of Mrs. VISMAYA A V

Place:

ADITH KUMAR T

Date:

ACKNOWLEDGEMENT

First of all I thank the Lord Almighty for his immense grace and blessings Showered on me at every Stages of This Work.

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ABSTRACT

The food management system will help the customers to get good quality and tasty food items from the nearby restaurants at their area. There are also providing food items to deliver at their home / requesting place as per customer's scheduled time. Customers can also verify the quality of Food / Bakery items on screening the review. Food management system is design to help the customers in finding the nearby restaurants / bakeries.

This system will also ensures the preservation of personal records and ensure the accessibility only by the authorized personal. To manage a food that include several modules. Admin is the main module to control all customers information , restaurant informations , bakery informations and its approval.

1. **Admin**: Admin should have great knowledge of the system. Admin is able to control the whole system. Admin can add, delete, update and modify the system and admin can also manages the information of the restaurant, bakery and customer and also can include new restaurants and bakeries.
2. **Common**: The common module is a portal module that allow users to type a user name and password to log in the system. Customer can log in by using login page and also include Restaurant registration , Customer registration and Bakery registration.
3. **Restaurant**: In this module it contains the details of food items, add food items, event

festivals, view orders received, view cancelled orders, view profile, manage food and change password. Restaurant can edit the personal details in this module.

4. **Customer**: Customer module allows to order restaurant and bakery for food items, view bakeries, restaurants, view list of items, change password, edit profile, add complaints if any, view my order placed and its status.

5. **Bakery**: In this module it contains the details of bakery items, add bakery items, manage bakery, view orders, view cancelled orders, view profile, change password.

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1.

INTRODUCTION

1.1.ABOUT THE PROJECT

The objective of this project titled “FOOD MANAGEMENT” is to develop a web based portal that is capable of managing the details of Restaurants , Bakery , Customer etc.. The purpose of the project is to build an application program to reduce the manual work for managing the Restaurants, Bakery.

The food management system will help the customers to get good quality and tasty food items from the nearby restaurants at their area. There are also providing food items to deliver at their home / requesting place as per customer’s scheduled time. Customers can also verify the quality of Food / Bakery items on screening the review. Food management system is design to help the customers in finding the nearby restaurants / bakeries.

This system will also ensures the preservation of personal records and ensure the accessibility only by the authorized personal. To manage a food that include several modules. Admin is the main module to control all customers information , restaurant informations , bakery informations and its approval.

2.

PROBLEM FORMULATION

1. OBJECTIVE OF THE SYSTEM

- Learn the basic elements of food management
- Learn the importance and benefits of the food
- Learn how to manage the details of food
- Manage the risk to build an application program to reduce the manual work for managing the food

3. SYSTEM ENVIRONMENT

3.1 HARDWARE SPECIFICATION

The selection of hardware configuration is a very important task related to the software development. Insufficient Random Access Memory may adversely on the speed and efficiency of the entire system. The processor should be powerful to handle the entire operations. The Hard disk should have sufficient capacity to store the file and applications.

Hardware Configuration Used:

Processor : Intel Pentium Dual Core

RAM : 1GB or Above

Hard Disk Drive : 40GB or Above

Speed : Above 1.5GHz

CPU : p3 or p4 etc

Monitor : Mono or color.

3.2 SOFTWARE SPECIFICATION

Operating system	: Windows 10
Front End	: .NET
Back End	: Microsoft SQL server

Operating system windows 10

One of the concerns of computer industry is ease of communication. We need to confront the human computer communication with a revolutionary approach. Programming has been activity that draws upon ability to think analytically and logically. So Graphical representations of programming aspects are preferable for human in computer communication.

There by we need an environment, that is, an interface that see, graphics to organize it's accomplish tasks. Such an environment is called Graphical User Interface (GUI).In Windows10, there are numerous features which have been added to automatically repair problems in applications and the operating system itself.

Windows 10 is the current release of Microsoft Windows, a series of operating systems produced by Microsoft for use on personal computers, including home and business desktops, laptops, net books, tablet PCs, and media center PCs. Windows 10 was released to manufacturing on July 22, 2009, and reached general retail availability worldwide on October 22, 2009, less than three years after the release of its predecessor, Windows Vista. Windows 7's server counterpart, Windows Server 2008 R2, was released at the same time.

Windows 10 includes a number of new features, such as advances in touch and handwriting recognition, support for virtual hard disks, improved performance on multi-core processors, improved boot performance, Direct Access, and kernel improvements. Windows 10 adds support for systems using multiple heterogeneous graphics cards from different

vendors (Heterogeneous Multi-adapter), a new version of Windows Media Center, a Gadget for Windows Media Center, improved media features, the XPS Essentials Pack and Windows Power Shell being included, and a redesigned Calculator with multiline capabilities including Programmer and Statistics modes along with unit conversion for length, weight, temperature, and several others.

For developers, Windows 10 includes a new networking API with support for building SOAP-based web services in native code (as opposed to .NET-based WCF web services), new features to shorten application install times, reduced UAC prompts, simplified development of installation packages, and improved globalization support through a new Extended Linguistic Services API. Windows 10 can boot the Windows system from a VHD; however, this ability is only available in the Enterprise and Ultimate editions.

Internet Spades, Internet Backgammon and Internet Checkers, which were removed from Windows Vista, were restored in Windows 10. Windows 10 includes Internet Explorer 8 and Windows Media Player 12. Users are also able to disable many more Windows components than was possible in Windows Vista.

The video modes supported in Windows 10 are 16-bit sRGB, 24-bit sRGB, 30-bit sRGB, 30-bit with extended color gamut sRGB, and 48-bit scRGB. Microsoft has also implemented better support for solid-state drives, including the new TRIM command, and Windows 10 is able to identify a solid-state drive uniquely.

3.3 FUNCTIONAL & NONFUNCTIONAL REQUIREMENT

Functional requirements

It consists of what task the system should perform, the processes involved, which data should the system holds and the Interfaces with the user. It has been mentioned as user requirements.

Nonfunctional requirements

It describes aspects of the system that are concerned with how the system provides the functional requirements.

They are:

- a. Security: Only users with valid password and username can login to view user's page.
- b. Performance and Response time: The system should have high performance rate when executing user's input and should be able to provide feedback or response within a short time. It also depends on the bandwidth of the connection used by the user and the network traffic.
- c. Error handling: Error should be considerably minimized and an appropriate error message that guides the user to recover from an error should be provided. Validation of user's input is highly essential.
- d. Availability: This system should always be available for access at 24 hours, 7 days a week.

SYSTEM ENVIRONMENT

(Front end)

4.5.1 About Programming Language

Microsoft.net

.NET is an integral part of many applications running on Windows and provides common functionality for those applications to run. Microsoft.NET is prefabricated infrastructure for solving common problems in internet applications. Microsoft.NET has been getting enormous amount of publicity lately, even for this industry Microsoft. NET is an add-on-run-time environment that runs on windows 2000 operating system. Later versions of .NET will probably be made part of the operating system, the U.S. department of justice willing.

Microsoft.net framework

The .NET Framework is Microsoft's comprehensive and consistent programming model for building applications that have visually stunning user experiences, seamless and secure communication, and the ability to model a range of business processes. The .NET Framework is a run-time environment that makes it much easier for programmers to write good, robust code quickly and to manage, deploy and revise the code. The programs and components that

you write execute inside this environment. It provides programmers with cool run-time features such as garbage collection and easier access to all system services. It adds many utility features such as easy internet and database access. It also provides a new mechanism for code reuse – easier to use and at the same time more powerful and flexible than COM. The .NET Framework is easier to deploy because it doesn't require registry settings.

Language(c#)

Microsoft Visual C# is a powerful but simple language aimed primarily at developers creating Applications by using the Microsoft .NET Framework. It inherits many of the best features of C++ and Microsoft Visual Basic, but few of the inconsistencies and anachronisms, resulting in a cleaner and more logical language. The advent of C# 3.0 has seen several important new features added to the language, including Generics, Iterators, and anonymous methods. By design, C# is the programming language that most directly reflects the underlying CLI. Most of its intrinsic types correspond to value-types implemented by the Common Language Infrastructure (CLI) framework. However, the language specification does not state the code generation requirements of the compiler: that is, it does not state that a C# compiler must target a Common Language Runtime, or generate Common Intermediate Language (CIL), or generate any other specific format. Theoretically, a C# compiler could generate machine code like traditional compilers of C++ or FORTRAN

Features of c#

. C# language is intended to be a simple, modern, general-purpose, object-oriented programming language.

. The language, and implementations thereof, should provide support for software engineering principles such as strong type checking, array bounds checking, detection of attempts to use uninitialized variables, and automatic garbage collection. Software robustness, durability, and programmer productivity are important.

. The language is intended for use in developing software components

suitable for deployment in distributed environments.

. Source code portability is very important, as is programmer portability, especially for those programmers already familiar with C and C++.

. Support for internationalization is very important.

. C# is intended to be suitable for writing applications for both hosted and embedded systems, ranging from the very large that use sophisticated operating systems, down to the very small having dedicated functions.

. Although C# applications are intended to be economical with regard to memory and processing power requirements, the language was not intended to compete directly on performance and size with C or assembly language

4.5.2 about Database (back end)

SQL SERVER 2008

SQL Server 2008 was released on August 6, 2008 and aims to make data management self-tuning, self organizing, and self maintaining with the development of SQL Server Always On technologies, to provide near-zero downtime. SQL Server 2008 also includes support for structured and semi-structured data, including digital media formats for pictures, audio, video and other multimedia data. In current versions, such multimedia data can be stored as BLOBs (binary large objects), but they are generic bit streams. Intrinsic awareness of multimedia data will allow specialized functions to be performed on them. According to Paul Flessner, senior Vice President, Server Applications, Microsoft Corp., SQL Server 2008 can be a data storage backend for different varieties of data: XML, email, time/calendar, file, document, spatial, etc as well as perform search, query, analysis, sharing, and synchronization across all data types.

SQL Server 2008 supports the ADO.NET Entity Framework and the reporting tools, replication, and data definition will be built around the Entity Data Model. SQL Server Reporting Services will gain charting capabilities from the integration of the data visualization products from Dundas Data Visualization, Inc., which was acquired by Microsoft.

On the management side, SQL Server 2008 includes the Declarative Management Framework which allows configuring policies and constraints, on the entire database or certain tables, declaratively. The version of SQL Server Management Studio included with SQL Server 2008 supports Intelligence for SQL queries against a SQL Server 2008 Database Engine. SQL Server 2008 also makes the databases available via Windows Power Shell providers and management functionality available as Cmdlets, so that the server and all the running instances can be managed from Windows Power Shell.

SQL Server includes better compression features, which also helps in improving scalability. It enhanced the indexing algorithms and introduced the notion of filtered indexes. It also includes Resource Governor that allows reserving resources for certain users or workflows.

Microsoft SQL Server also allows user-defined composite types (UDTs) to be defined and used. It also makes server statistics available as virtual tables and views (called Dynamic Management Views or DMVs). In addition to tables, a database can also contain other objects including views, stored procedures, indexes and constraints, along with a transaction log.

SQL Server buffers pages in RAM to minimize disc I/O. Any 8 KB page can be buffered in-memory, and the set of all pages currently buffered is called the buffer cache. The amount of memory available to SQL Server decides how many pages will be cached in memory. The buffer cache is managed by the Buffer Manager.

SQL Server ensures that any change to the data is ACID-compliant, i.e. it uses transactions to ensure that the database will always revert to a known consistent state on failure. Each transaction may consist of multiple SQL statements all of which will only make a permanent change to the database if the last statement in the transaction (a

COMMIT statement) completes successfully. If the COMMIT successfully completes the transaction is safely on disk.

SQL Server allows multiple clients to use the same database concurrently. As such, it needs to control concurrent access to shared data, to ensure data integrity when multiple clients update the same data, or clients attempt to read data that is in the process of being changed by another client. SQL Server also provides the optimistic concurrency control mechanism, which is similar to the multi version concurrency control used in other databases.

4. SYSTEM ANALYSIS

4.1 REQUIREMENT ANALYSIS

Requirement analysis provides the software designer with representation of information, function and behavior that can be translated to architectural, interface and component – level designs. Finally the analysis model and the requirement specification provide the development and the customs with the means to access quality once software is built.

Requirements analysis can be a long and arduous process during which many delicate psychological skills are involved. New systems change the environment and relationships between people, so it is important to identify all the stakeholders, take into account all their needs and ensure they understand the implications of the new systems. Analysts can employ several techniques to elicit the requirements from the customer. **Requirements analysis** in systems engineering and software engineering, encompasses those tasks that go into determining the needs or conditions to meet for a new or altered product, taking account of the possibly conflicting requirements of the various stakeholders, such as beneficiaries or user. Requirements analysis is critical to the success of a systems or software project. The

requirements should be documented, actionable, measurable, testable, traceable, related to identified business needs or opportunities, and defined to a level of detail sufficient for system design.

Requirement of this project is to make the manual procedures in toll booths to be computerized. This project aims at making a system which will make more effective and time consuming for Toll booths handling. The system will identify the details of vehicles and owner and proper functioning should be worked. The nature of the manual work is so tedious, the idea of future growth under this system is frightening. In order to overcome the problems faced by the existing system the proposed system was developed.

4.2 Existing System

In the present scenario, people have to physically visit the hotels or restaurants for eating food and have to make payments through cash mode most of the time due to unawareness of advanced technologies at certain places. In this method time as well as physical work is required, among which time is something that no one has in ample amount.

The traditional food ordering procedure is not efficient enough for hotels and restaurants, as they have to deal with the crowd, in their restaurant. The old methods can be classified into categories which are paper grounded and verbal grounded. For paper-based work, the waiter comes and pens down foods that customers order and pass the food list containing paper to the chefs or cooks in the kitchen for further process.

Disadvantages Of Existing System

- Mistakes are made when taking the orders of the customers

- The process of collecting customers' purchases order is very tedious. This makes it impossible to deliver goods on time.
- It leads to lack of understanding between the customers and the employees.
- The record keeping system is poor. Losses of vital records have been reported in the past consequently. Besides, protecting the file system from unauthorized access is a problem that has defiled solution (php project centers in chennai).
- It causes reduction of production flow.
- In the existing system, the person has to visit nearby hotels and bakery for buying food items. Or they need to call particular shops for enquiry and ordering.
- As there are some apps exist for online food ordering, but some times no delivery is available for particular area.

4.3 Proposed System

The proposed system is developed to manage ordering activities in fast food. It helps to record customer submitted orders. The system should cover the following functions in order to support the admin business process for achieving the objectives:

ADVANTAGES:

- It allows the customer to make order, view order and make changes before submitting their order and allow them make payment through prepayment card or credit card or debit card. No-need to go shop and select product.
- It provides interface that allows promotion and menu.

- It prevent interface that shows customers' orders detail to front-end and kitchen staffs for delivering customers' orders It is not a time-consuming process.
- Tools that generate reports that can be used for decision making.
- A tool that allows the management to modify the food information such as price, add a new menu and many others as well as tools for managing user, system menu and promotion records.
- Accuracy in handling of data.
- Better storage and faster retrieval system.

5. FEASIBILITY STUDY

Feasibility analysis is the procedure for identifying the candidate system, evaluating and electing the most feasible system. This is done by investigating the existing system in the area under investigation or generally ideas about a new system. It is a test of a system proposal according to its workability, impact on the organization, ability to meet user needs, and effective use of resources. The objective of feasibility study is not to solve the problem but to acquire a sense of its scope.

Feasibility analysis involves 8 steps:

1. Form a project team and appoint a project leader.
2. Prepare system flow charts.

3. Enumerate potential candidate system.
4. Describe and identify characteristics of candidate systems.
5. Determine and evaluate performance and cost effectiveness of each candidate system.
6. Weigh system performance and cost data.
7. Select the best candidate system.
8. Repair and report final project directive to management.

Three key considerations are involved in the feasibility analysis: economic, technical and behavioral.

- **5.1 Technical Feasibility**

The assessments of technical feasibility centers on the existing system and to what extent it can support the proposed addition. This was based on an outline design of system requirements in turns of inputs, files, programs, procedures, and staff. It involves financial considerations to accommodate technical enhancements.

- **5.2 Economical Feasibility**

Economic analysis is the most frequently used method for evaluating the effectiveness of a candidate system. It is more commonly known as cost benefit analysis, the procedure to determine the benefits and saving that are expected from a candidate system and compare them with costs. If the benefits outweigh costs then a decision is made to design and implement the system. Otherwise make alterations in the proposed system.

- **5.3 Behavioral Feasibility**

People are inherently resistant to change, and computers have been known to facilitate change. An estimate should be made about the reaction of the user staff towards the development of a computerized system. Computer installations have something to do with turnover, transfers and changes in job status. The introduction of a candidate system requires special effort to educate, sell and train the staff for conducting the business. The candidate system was found to be technically, economically, and behaviorally feasible. The system was developed user friendly, needless training and improves the working environment. Justification for any capital outlay is that it will increase profit, reduce expenditure or improve the quality of service or goods, which in turn may be expected to provide increased profits. Disregarding the initial expenses, the candidate system was assessed to be feasible in all ways.

6. PROJECT DESCRIPTION

6.1 PROBLEM DEFINITION

A web based portal for food management including customer registration, bakery registration, restaurant registration, view bakeries, restaurants etc..

6.2 OVERVIEW OF THE PROJECT

The system should provide login, profile updation, view restaurants, view bakery to order food items, test result view complaints, add complaints for the customers and logout.

The system should provide login, view user information, add categories and subcategories, delete categories and subcategories, view complaints for admin.

6.3 MODULE DESCRIPTION

The administrator has the ultimate control over the entire Clinic Management System. Admin has the ability to identify the user and their information, view complaints, add categories and subcategories etc..

User module deals with the process of registering to the portal, order food item, updating the profile, submitting complaints.

Modules

1. Admin Module:

➤ Admin Login

- Add bakery item
- Add catagory
- Add district
- Add item
- Add location
- Approval bakery
- Manage bakery
- Manage customer
- Manage restaurant
- Manage rejected bakery
- Manage rejected restaurant
- Restaurant approval
- View complaints
- Home

2. Common:

- About us
- Contact us
- Login
- Home
- Index

- Gallery
- Restaurant registration
- Bakery registration
- Customer registration
- Logout

3. Bakery:

- Home
- Add bakery item
- Change password
- Manage bakery
- View order
- View cancelled order
- View profile

4. Restaurant:

- Change password
- Add restaurant item
- Home page
- View profile
- Manage food
- Event festivals
- View order

5. Customer

- Add Complaint
- List Bakery Items
- Change Password
- Restaurant Menu
- Edit Profile
- Event Festival
- Order Bakery
- Order Restaurant
- View Bakery
- View My Order
- View Restaurant

7. SYSTEM DESIGN

System design is the process or art of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. System design involves designing a new system or computer application that will meet the requirements identified during system analysis. System depending designing is concerned with the computer oriented design of the system-details of the input transactions, the details of the printed reports, screens and other outputs ,database structure ,the contents of records, the processing required and the efficiency of the system from a computer processing point of view. In general the following design objectives must be kept in mind:

- Practicality of the System
- Efficiency of the System
- Minimized cost of the system
- Flexibility of the System
- Security of the System

The system design process is to achieve objectives like practicability, efficiency, minimized cost, flexibility, and security of the system. It involves, first, logical design and then, physical construction of the system. The logical design specifies the structure and characteristics of output, input, files, database and procedures. The physical design involves the construction of actual program files and a working system.

7.1 INPUT DESIGN

Input design plays a vital role in the life cycle of the software development. It requires the very careful attention of the developers. It specifies the manner in which the data enters the system for processing. Input design can ensure the reliability of the system and the producer's results from the accurate data or it may result in the output of an enormous data. According to the software engineering concepts, the input forms or screens are designed to provide to have a validation control over the input limit.

This system has input screens in almost all the modules. Error messages have been developed to alert the user whenever he commits some mistakes and guides him in the right way. Input design determines whether the user interacts with the system efficiently.

It consists of developing specifications and procedures for the data preparations. So, structured steps are necessary to put the transaction data into usable form for processing.

Input design is the process of converting the user-originated inputs to a computer-based format. It also includes the method of input and entry into the system. There can be many factors that can be taken into account

- Flexibility of format
- Speed
- Accuracy
- Type of input
- Verification methods
- Offline contact facility
- Automatic features
- Ease of use

Keyboard is the most commonly used input media. In accurate, input data are the most common cause of errors in data processing. Errors entered by the user can be controlled by the input design.

The design for handling input specifies how the data are accepted for computer processing. Input design is a part of the overall system design that needs careful attention and it includes specifying the means by which actions are taken.

A system user interacting through a workstation must be able to tell the system whether to accept input produced a report or end the processing. The collection of the input data is considered to be the most extensive part of the system design.

Since the inputs have to be planned in such a manner so as to get the relevant information, extreme care has to be taken to obtain the information. If the data going into the system incorrect, then the processing and the outputs will magnify these errors.

7.2 OUTPUT DESIGN

Often the most important feature of the information system for users is the output it produces. Without quality, outputs we may even fill the entire system is so unnecessary that they avoid using it and that possibly causes the system fail.

Output design refers to the generation of the results and information of the project. It should be an organized and well out manner. It involves conceiving, planning out and specifying the external observable characteristics of the product. It includes user displays and the report formats.

The normal procedure is to design the outputs in detail first and then to work back with the inputs. The outputs can be in the form of the operational documents and report. The input records have to be validated, edited, organized and accepted by the system before being processed to produce the output.

Output is the most important and direct source of information to the users. Designing the computer output should produce in an organized manner. The right output must be developed while ensuring that each output element is designed so that people will find the system easy for the use or user friendly.

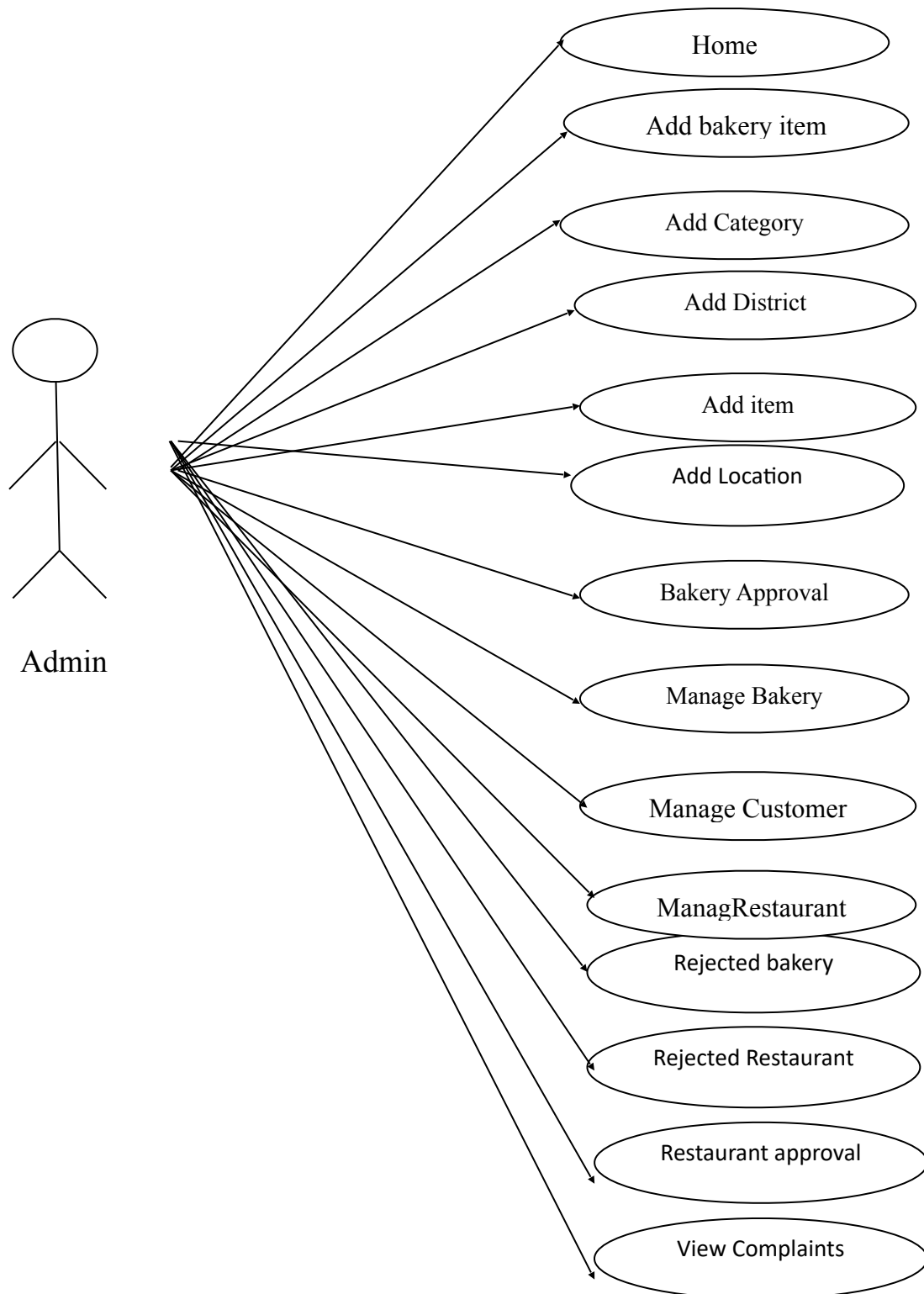
The output design has to be done so that the results of the processing should be communicated to the user. Effective output design will improve the clarity and performance of

the outputs. Output is the main reason for developing the system and the basis on which the usefulness of the application will be evaluated. Output design phase of the system is concerned with the convergence of the information to the end user in a user friendly manner. The output design should be efficient, intelligible so that the system relationship with the end user improves and thereby enhancing the process of the decision making.

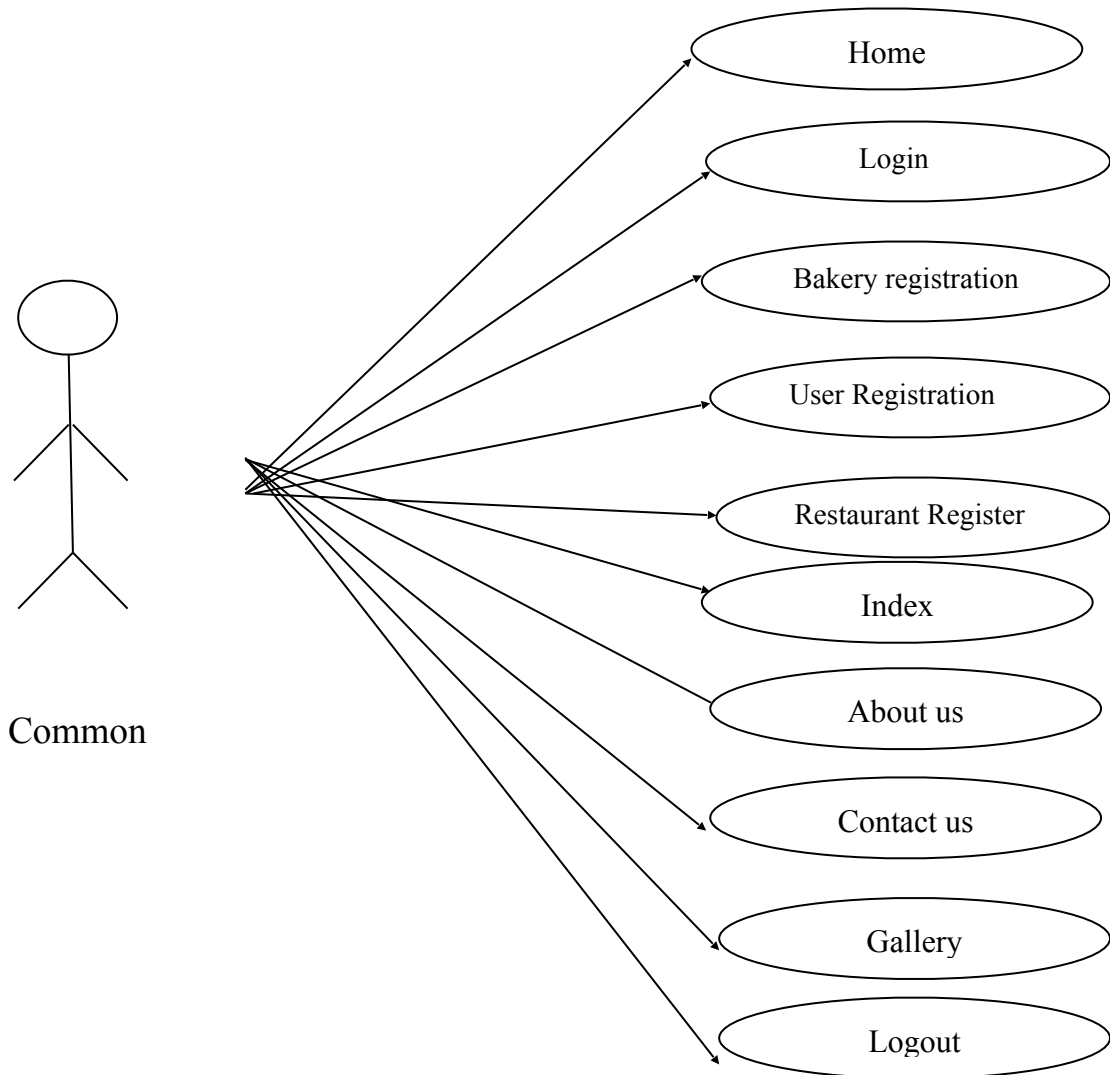
7.3 USE CASE DIAGRAMS

A Use case is a description of set of sequence of actions. Graphically it is rendered as an ellipse with solid line including only its name. Use case diagram is a behavioral diagram that shows a set of use cases and actors and their relationship. It is an association between the use cases and actors. An actor represents a real-world object.

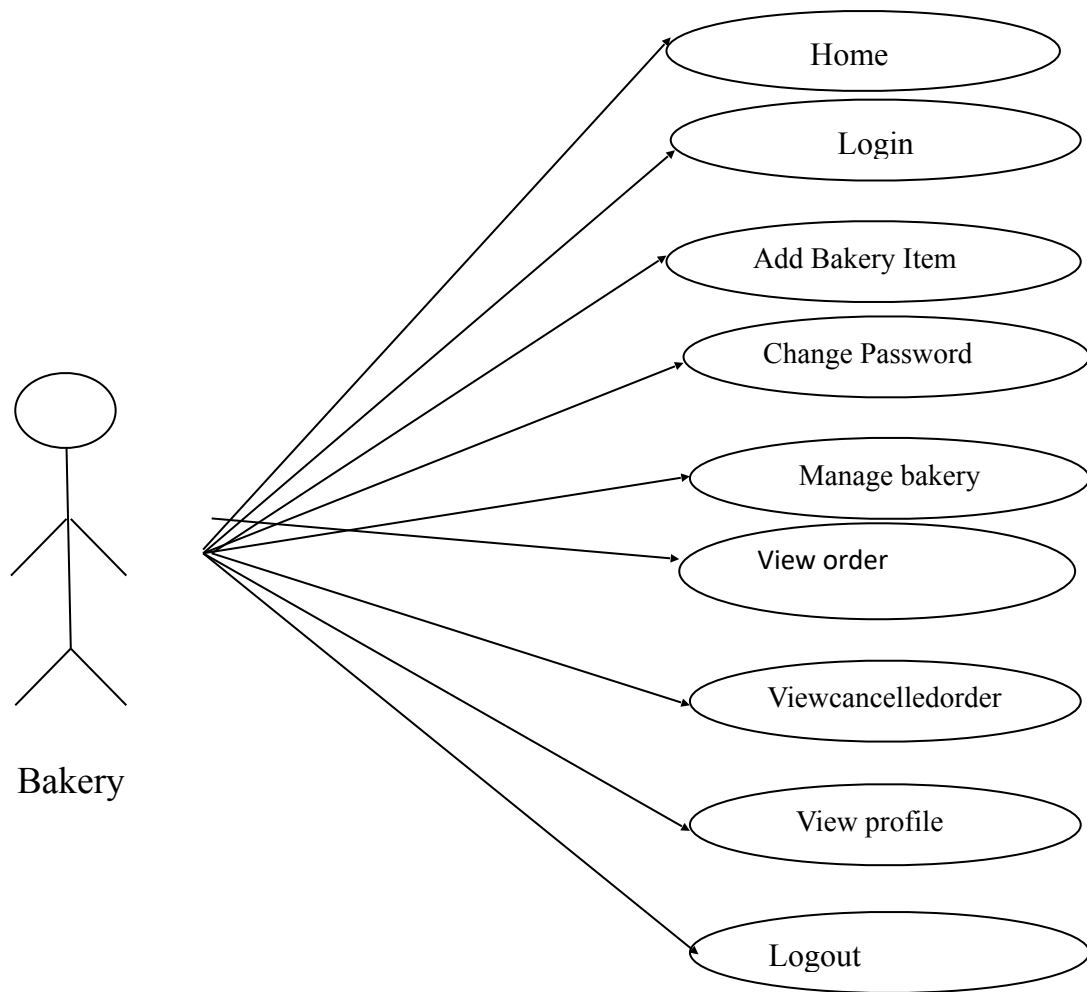
Use Case For Admin



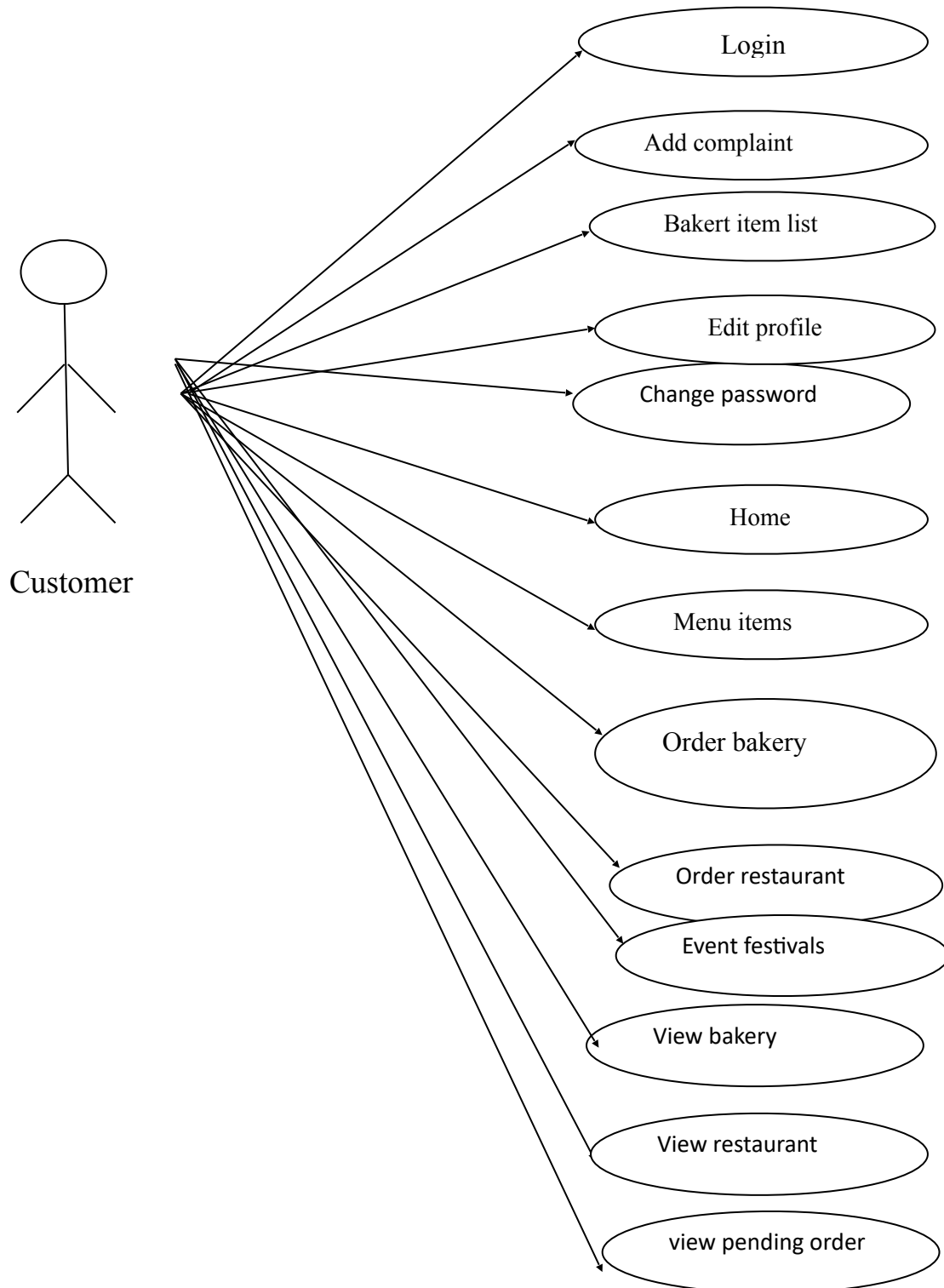
Use Case For Common



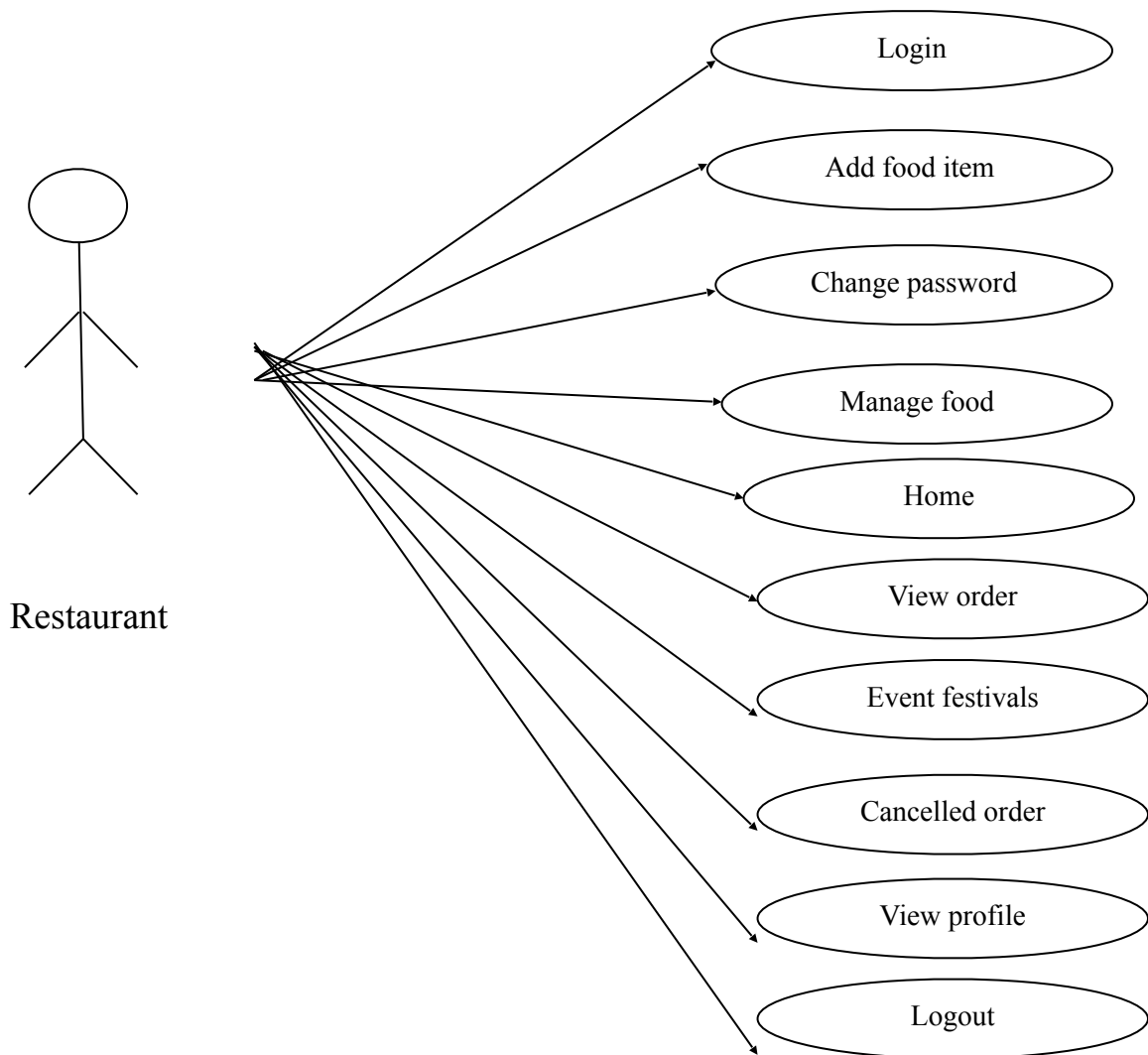
Use Case For Bakery



Use Case For Customer



Use Case For Restaurant



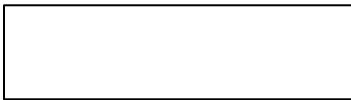
7.4 E-R DIAGRAM

An entity-relationship diagram is a data modeling technique that creates a graphical representation of the entities, and the relationship between the entities, within an information system.

There are three basic elements in ER model:

- Entities are the “things” about which we seek information.
- Attributes are the data we collect about the entities.
- Relationships provide the structure needed to draw information from multiple entities.

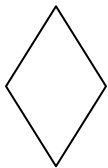
ER diagram symbols :



Entity



Attributes

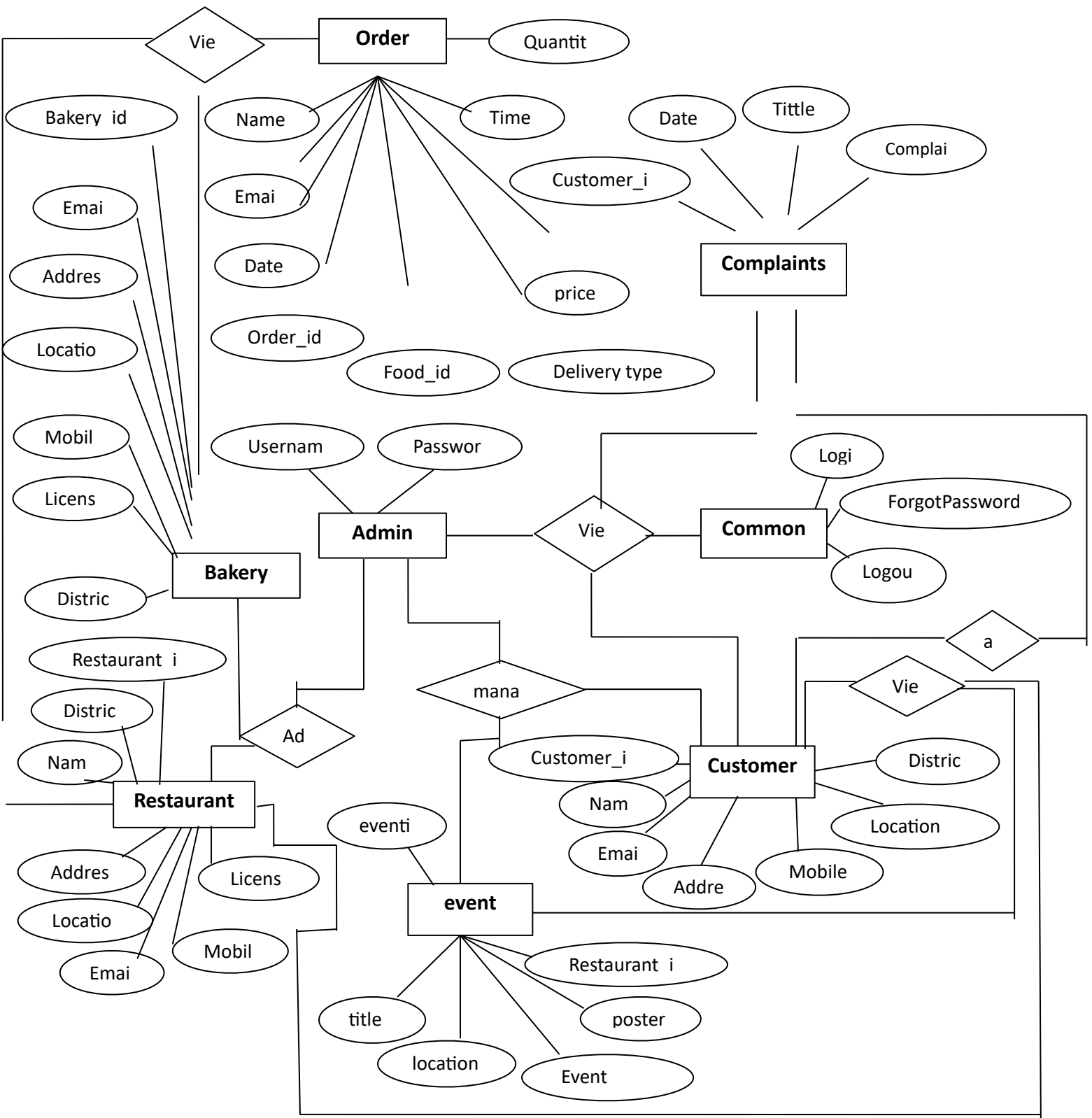


Relation



Connection

E-R DIAGRAM



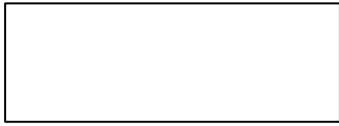
7.5 DATA FLOW DIAGRAMS

A data flow diagram (DFD) or a bubble chart is a graphical tool for structured analysis, DFD models a system by using external entities from which data flows to a process, which transforms the data and creates output-data-flows, which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

DFDs can be hierarchically organized, which help in partitioning and analyzing large systems. As a first step, one dataflow diagram can depict an entire system which gives the system overview. It is called context diagram of level 0 DFD. The context diagram can be further expanded.

The successive expansion of a DFD from the context diagram to those giving more details is known as leveling of DFD .Thus a top down approach is used, starting with an overview and then working out the details. The main merit of DFD is that it can provide an overview of what data a system would process, what transformation of data are done, what files are used, and where the results flow ,the data flow has been represented as a hierarchical DFD .Context Level DFD was drawn first. Then the processes were decomposed into several elementary levels and were represented in the order of importance.

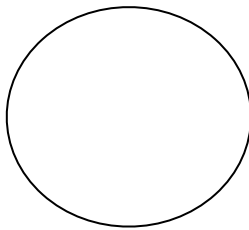
Basic DFD Diagrams



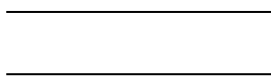
Represent source or destination of data. It is considered to be outside the context of data flow model.



Represents data flow

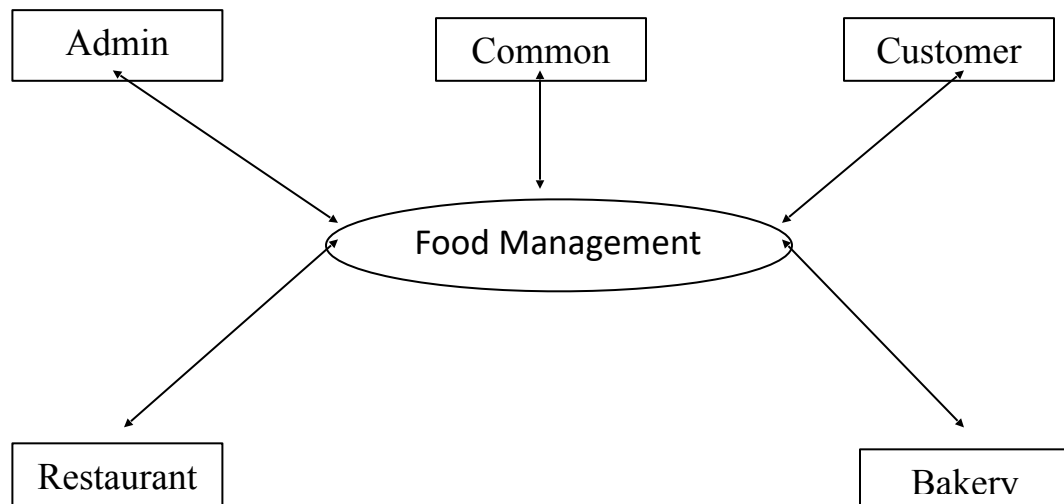


A process that transforms incoming data into outgoing flow.



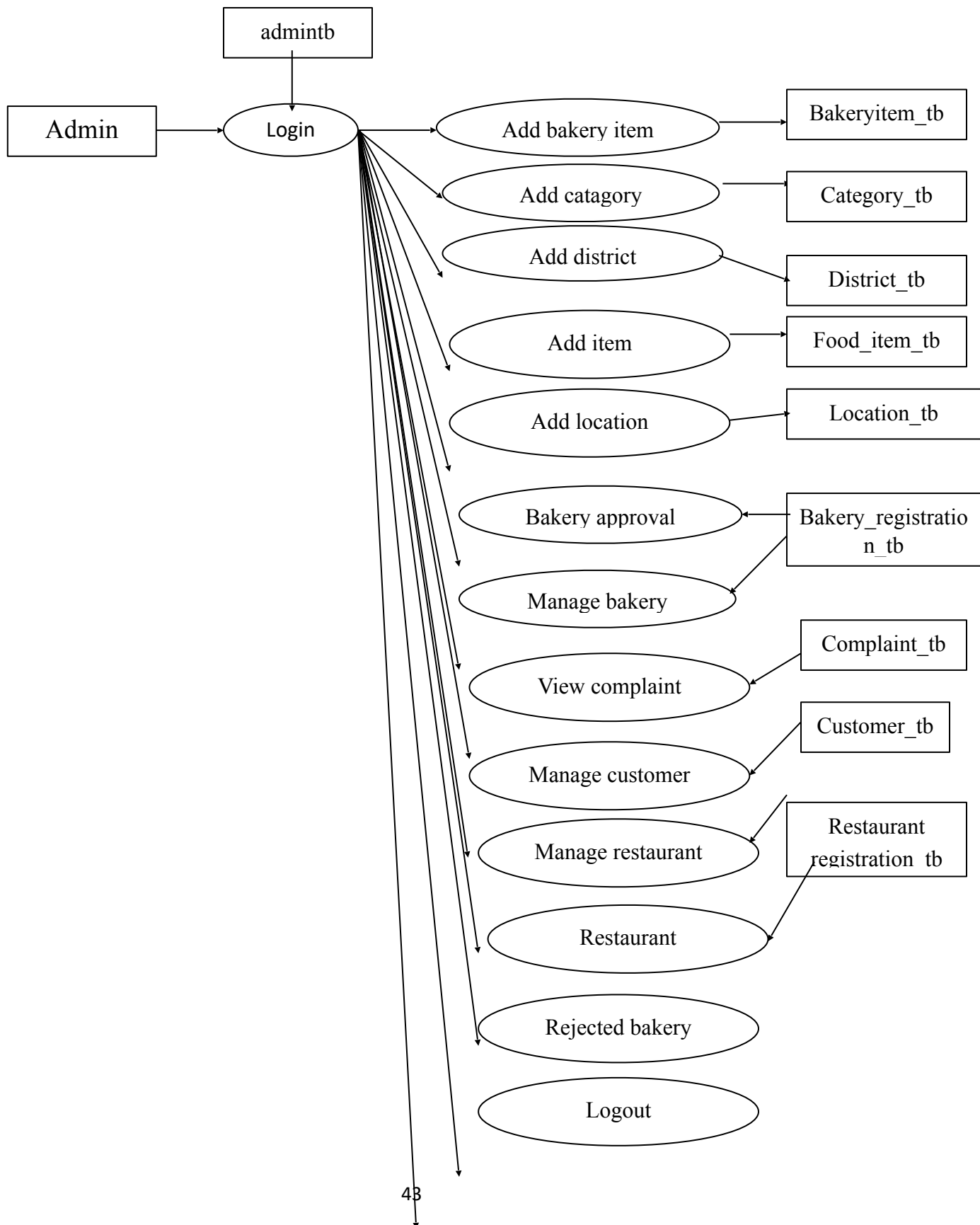
Data store – repository of data that is to be stored for use by one or more process. It may be simple as buffer or queue or sophisticated as relational database.

Level 0:

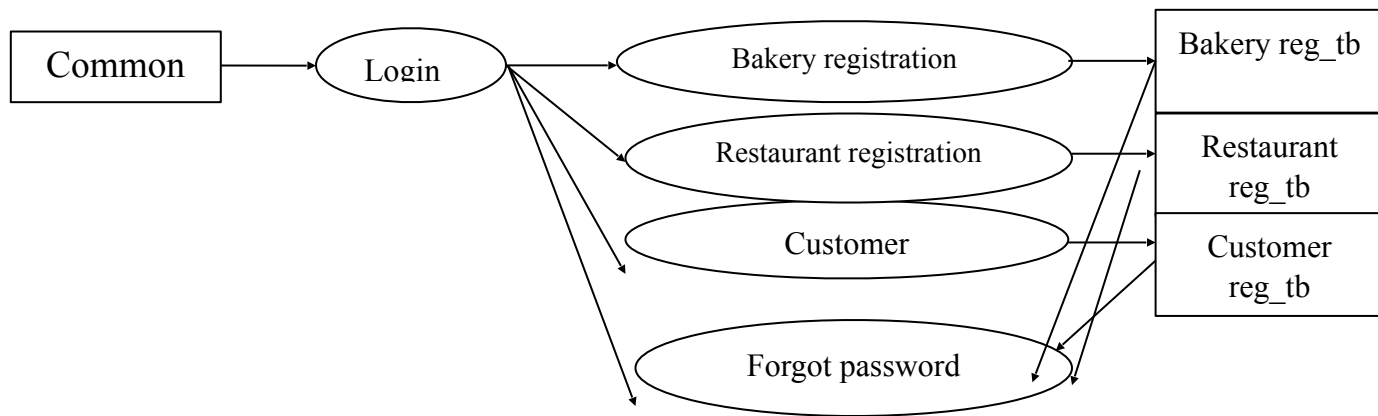


Level 1:

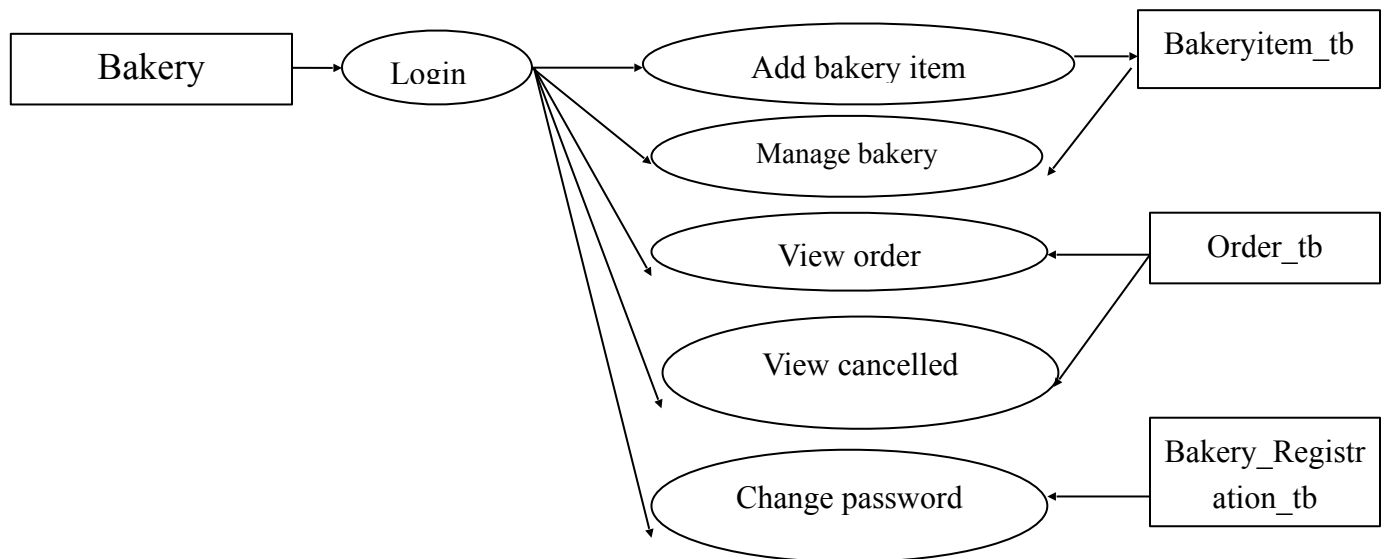
Admin



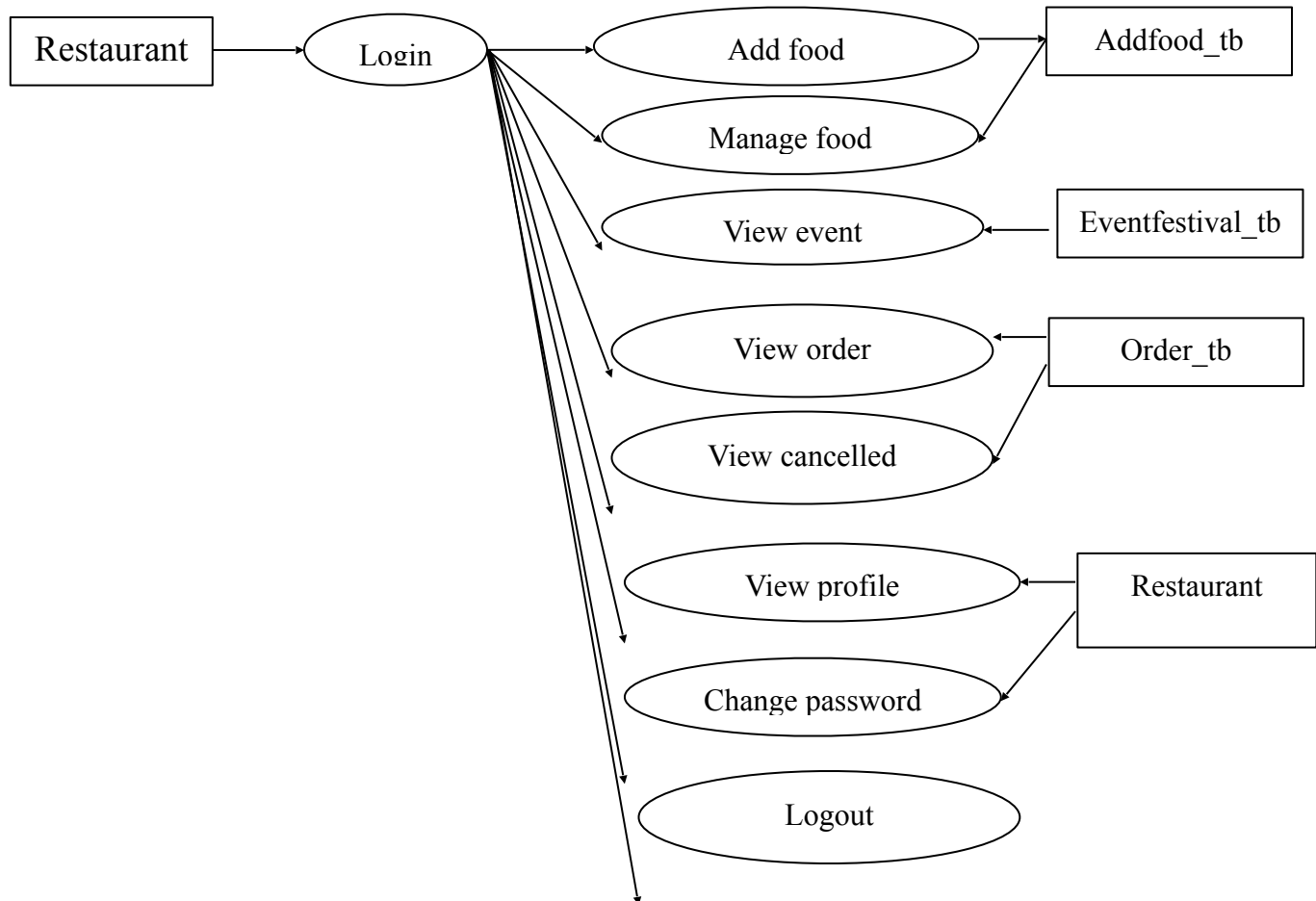
Common



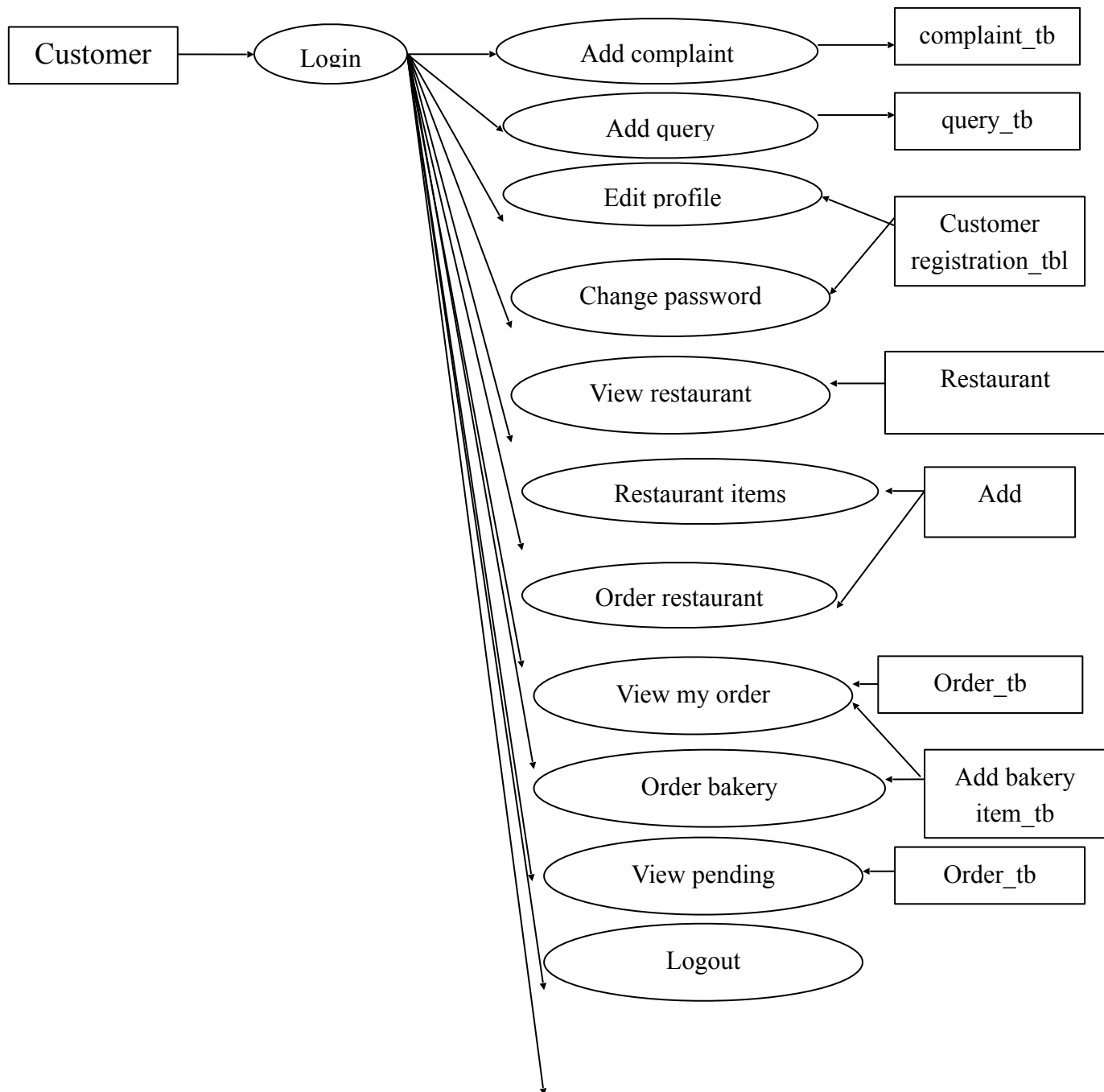
Bakery



Restaurant



Customer



7.6 DATABASE DESIGN

The data base design is a logical development in the methods used by the computer to access and manipulate data stored in the various parts of the computer system. Database is defined as an integrated collection of data. The overall objective in the development of database technology has been to treat data as organization recourses and as an integrated while. The main objectives of database are data integration, data integrity and data independence.

Database management system (DBMS) allows the data to be protected and organized separately from other recourses like hardware, software and programs. DBMS is a software package, which contains components that are not found in other data management packages. The significance of DBMS is the separation of data as seen by the programs and data as stored on the direct access storage devices. I.e. the difference between the logical and physical data.

Database design is the first step of the three design activities that are conducted during software engineering process. The impact of data structure on program structure and procedural complexity causes data design to have a profound influence on software quality.

The concepts of shared memory and semaphores provide the foundation for an approach to data design. The database is structured in fixed format records of several types. Each record type defines a fixed number of fields or attributes and each field is usually of a fixed length. Usually centralized data management software is called RDBMS.

The project uses the following tables:

1. Add_Bakery_Item_tb
2. Add_District
3. Add_Food_tb
4. Admin_tb
5. Bakery_Registration_tb
6. Bakery_Types_tb
7. Category_tb
8. Complaint_tb
9. Customer_tb
10. Event_Festival
11. FoodTypes_tb
12. Location_tb
13. Order_tb
14. Restaurant_registration_tbl

Table 7.7.1 – Add_bakery_item_tb

	Column Name	Data Type	Allow Nulls
►	Id	nchar(10)	☒
	Photo	varchar(MAX)	☒
	Item_Name	varchar(50)	☒
	Catagory	varchar(50)	☒
	Type	varchar(50)	☒
	Rate	varchar(50)	☒
	Bakery_id	varchar(50)	☒

Table 7.7.2 – Add_district

	Column Name	Data Type	Allow Nulls
▶	D_id	varchar(MAX)	☒
	D_name	varchar(50)	☒

Table 7.7.3 – Add_Food_tb

	Column Name	Data Type	Allow Nulls
▶	id	nchar(10)	☒
	Photo	varchar(MAX)	☒
	Item_Name	varchar(50)	☒
	Catagory	varchar(50)	☒
	Type	varchar(50)	☒
	Description	varchar(MAX)	☒
	Rate	varchar(50)	☒
	Resturant_id	nchar(10)	☒

Table 7.7.4 – Admin_tb

	Column Name	Data Type	Allow Nulls
▶	Admin_id	nchar(10)	☒
	Username	varchar(50)	☒
	Password	varchar(50)	☒

Table 7.7.5 – Bakery_Registration_tb

	Column Name	Data Type	Allow Nulls
▶	Bakery_id	varchar(50)	☒
	Bakery_name	varchar(MAX)	☒
	Address	varchar(MAX)	☒
	District	varchar(50)	☒
	Location	varchar(MAX)	☒
	Email	varchar(MAX)	☒
	Mobileno	varchar(50)	☒
	Licence	varchar(MAX)	☒
	Username	varchar(MAX)	☒
	Password	varchar(MAX)	☒
	Status	varchar(MAX)	☒

Table 7.7.6 – Bakery_Types_tb

	Column Name	Data Type	Allow Nulls
▶	Type_id	varchar(50)	☒
	Type_name	varchar(50)	☒
	Catagory_id	varchar(50)	☒

Table 7.7.7 – Category_tb

	Column Name	Data Type	Allow Nulls
▶	Cid	varchar(50)	☒
	Catagory	varchar(50)	☒

Table 7.7.8 – Complaint_tb

	Column Name	Data Type	Allow Nulls
▶	Cid	varchar(50)	☒
	Tittle	varchar(MAX)	☒
	Complaint	varchar(MAX)	☒
	Date	varchar(MAX)	☒
	Customer_id	varchar(50)	☒

Table 7.7.9 – Customer_tb

	Column Name	Data Type	Allow Nulls
►	Cid	varchar(50)	☒
	Name	varchar(50)	☒
	Address	varchar(MAX)	☒
	District	varchar(50)	☒
	Location	varchar(50)	☒
	Mobileno	varchar(50)	☒
	Email	varchar(50)	☒
	Username	varchar(50)	☒
	Password	varchar(50)	☒

Table 7.7.10 – Event_Festival

	Column Name	Data Type	Allow Nulls
►	Event_id	varchar(50)	☒
	Event_Title	varchar(MAX)	☒
	Event_Description	varchar(MAX)	☒
	Location	varchar(MAX)	☒
	Landmark	varchar(MAX)	☒
	Starting_Date	varchar(MAX)	☒
	Ending_Date	varchar(MAX)	☒
	Poster	varchar(MAX)	☒
	Restaurant_id	varchar(50)	☒

Table 7.7.11 – Food_Types_tb

	Column Name	Data Type	Allow Nulls
►	Type_id	varchar(50)	☒
	Type_name	varchar(50)	☒
	Catagory_id	varchar(50)	☒

Table 7.7.12 – Location_tb

	Column Name	Data Type	Allow Nulls
►	Lid	varchar(50)	☒
	Lname	varchar(50)	☒
	Did	varchar(50)	☒

Table 7.7.13 – Order_tb

	Column Name	Data Type	Allow Nulls
►	Order_id	varchar(MAX)	☒
	Food_id	varchar(MAX)	☒
	Customer_id	varchar(50)	☒
	Quantity	varchar(MAX)	☒
	Total_price	varchar(MAX)	☒
	Name	varchar(50)	☒
	Address	varchar(MAX)	☒
	Landmark	varchar(50)	☒
	Mobileno	varchar(50)	☒
	Email	varchar(50)	☒
	Delivery_type	varchar(50)	☒
	Date	varchar(MAX)	☒
	Time	varchar(MAX)	☒
	Payment_method	varchar(MAX)	☒
	Status	varchar(50)	☒

Table 7.7.14 – Restaurant_Registration_tb

	Column Name	Data Type	Allow Nulls
►	Rstid	varchar(50)	☐
	Rstname	varchar(50)	☒
	Address	varchar(50)	☒
	District	varchar(50)	☒
	Location	varchar(50)	☒
	Email	varchar(50)	☒
	Mobileno	varchar(50)	☒
	Licence	varchar(MAX)	☒
	Username	varchar(50)	☒
	Password	varchar(50)	☒
	Status	varchar(50)	☒

8. SYSTEM TESTING

Testing is the process of executing a program with the intent of finding any errors. Testing is vital to the success of the system. Without proper testing, hidden errors will surface after some time of use and perhaps irreversible damage has been done to valuable data. A series of tests like responsiveness, its value, stress and security are performed before the system is ready for user acceptance testing. System testing follows the logical conclusion that all the parts of the system are tested and found to be working properly under all kinds of situations, and then the system is achieving its goal of processing the data perfectly according to the user rules and requirements.

System testing is aimed at ensuring the system works accurately before live operation commences. Testing is vital to the system. System testing makes a logical assumption that if all parts of the system are correct, the goal will be successfully achieved. The candidate system is subjected to a variety of tests: Online response, Volume, Stress Recovery & Security and Usable tests. A series of testing are performed for the proposed system before the system is ready for user acceptance testing. Nothing is complete without, as it is vital success of the system.

8.1 TESTING ISSUES

Unit Testing:

Unit testing focuses verification effort on the smallest unit of the software designs the module. To check whether each module in the software works properly so that it gives desired outputs to the given inputs. All the validations and conditions are tested in the module level the unit test. Boundary condition is tested that the modules operate at boundaries. All independent paths through the control structure ensure that all statements in a module have been executed at least once.

Integration Testing:

The major concerns of integration testing are developing incremental strategies that will limit the complexity of entire actions among components as they are added to the system. Developing a component as they are added to the system, developing an implementation & integration schedules that will make the modules available when needed, designing test cases that will demonstrate the viability of evolving system. Though each program works individually, they should work after linking them together. This is also referred to as Interfacing. Data may be lost across interface and one module can have an adverse effect on another. Subroutines, after linking, may not be do the desired function expected by the main routine. Integration testing is a systematic technique for constructing program structure while at the same time, conducting test to uncover errors associated with the interface. In the testing, the programs are constructed and tested in small segments.

System Testing:

When a system is developed, it is hoped that it performed properly. In practice, however, some errors always occur. The main purpose of testing an information system is to find the errors and correct them. A successful test is one, which finds an error.

The main objectives of system testing are

- To ensure during operation the system will perform as per specification.
- To make sure that the system meets user's requirement during operation.
- To verify that the controls incorporated in the system function as intended.
- To see that when correct inputs are fed to the system the outputs are correct.
- To make sure that during operation, incorrect input and output will be deleted.

The scope of a system test should include both manual operations computerized. Operations system testing is a comprehensive evaluation of the programs, manual procedures, computer operations and controls. System testing is the process of checking if the developed

system is working according to the original objectives and requirements. All testing needs to be conducted in accordance tot the test conditions specified earlier.

Data Validation Testing:

Data validation testing is done to see whether the corresponding entries made in the tables are correct. Proper validations are done in case of insertion and updating of the tables. If any such case arises, then proper error messages or warning, if any, has to be displayed.

The different test cases are: Guarantee that all independent parts within a module have been exercised at least once.

- Exercise all logical decision on their true/false side.
- Exercise all loops at their boundaries and within their operational bounds.
- Exercise internal data structure to ensure their validity.
- Each module was tested and the tested modules were linked and integration test was carried out.

9.

SYSTEM IMPLEMENTATION

Implementation is a stage where theoretical design is turned to the working system. The implementation phase is used to test the developed package with sample data, correcting the error identified, appearing the user of the various special facilities and features of the computerized system. It also involves user training for minimizing resistance to change and giving the new system a change to prove its worth. The successful implementation of the new system depends upon the involvement of the user. Implementation means converting a new or revised system design into an operational one. The implementation includes all those activities that take place to convert from the old system to the new one. There are several activities involved while implementing a project:

- Careful planning.
- Investigating the current system and its constraints on implementation.
- Design of methods to achieve the changeover.
- Training of the staff in the changeover procedure and evaluation of change over method.

Implementation is the final stage and it is an importance phase. The first task in implementation was the implementation planning, that is deciding on methods to be adopted. After the system was implemented successfully, training of the user was one of the most important subtasks of the developer. For this purpose the user or system manual were prepared and handled over to the user to operate the developed system. So change over plays a vital role, which checks the developed tool for the following requirements, and then only the user accepted the developed tool. The change over took place only when the system had been proved to the satisfaction of the system analysis and other implementation activities have been completed.

9.1 IMPLEMENTATION METHODS:

There are several methods for handling the implementation and consists for changing from the old to the new computerized system. The most secure method for conversion from the old system is to run the old and new system in parallel .In this approach; a person may operate in the manual processing system as well as start operating the new computerized system.

Another commonly used method is a direct cut over the existing manual system to the computerized system. The change may be within a week or a day. This strategy requires planning. A working version of the system can also be implemented in one part of the organization and the changes can be made as and when required, but this method is less preference due to the loss of entire system. After the system is implemented, a review should be conducted to determine whether the system is meeting expectations where improvements are needed.

9.2 IMPLEMENTATION PLAN:

Implementation plan includes a description of all activities that must occur to implement the new system and to put into operation. It defines the personnel responsible for the activities and prepares a time chart for Implementation of the system. The Implementation plan should anticipate possible problems and must be able to deal with them. The usual problem may be missing documents, missed data formats between current and new files, errors in data translation, missing data etc.

9.3 USER TRAINING:

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. Even well designed and technically elegant systems can succeed or fail, because of the way they are operated and used. For this purpose user manuals are prepared and handed over to the user to operate the Developed system. Thus the users are trained to operate the developed system. Both the hardware and software securities are made to

run the developed systems successfully in future. In order to put new application system into use, the following activities were taken care of:

- Preparation of user and system documentation
- Conducting user training with demo and hands on
- Test run for some period to ensure smooth switching over the system

The users are trained to use the newly developed functions. User manuals describing the procedures for using the functions listed on menu and circulated to all the users. It is confirmed that the system is implemented up to users need and expectations.

9.4 IMPLEMENTATION:

Admin, User

Admin

Add category

Add Sub category

View category

View Sub category

View request

View users

View Feedback

User

View category

View Sub category

Login

Order product

Edit Profile

Change Password

View wishlist

Send Feedback

10. SYSTEM MAINTENANCE

Software Development has many phases. These phases include Requirements Engineering, Architecting, Design, Implementation, Testing, Software Deployment, and Maintenance. Maintenance is the last stage of the software life cycle. After the product has been released, the maintenance phase keeps the software up to date with environment changes and changing user requirements. The earlier phases should be done so that the product is easily maintainable. The design phase should plan the structure in a way that can be easily altered. Similarly, the Software maintenance is the modification of a software product after delivery to correct faults, to improve performance or other attributes, or to adapt the product to a modified environment implementation phase should create code that can be easily read, understood, and changed. Maintenance can only happen efficiently if the earlier phases are done properly.

There are four major problems that can slow down the maintenance process: unstructured code, maintenance programmers having insufficient knowledge of the system, documentation being absent, out of date, or at best insufficient, and software maintenance having a bad image. The success of the maintenance phase relies on these problems being fixed earlier in the life cycle.

Maintenance consists of four parts. Corrective maintenance deals with fixing bugs in the code. Adaptive maintenance deals with adapting the software to new environments. Perfective maintenance deals with updating the software according to changes in user requirements. Finally, preventive maintenance deals with updating documentation and making the software more maintainable. All changes to the system can be characterized by these four types of

maintenance. Corrective maintenance is ‘traditional maintenance’ while the other types are considered as ‘software evolution.’

11. CONCLUSION

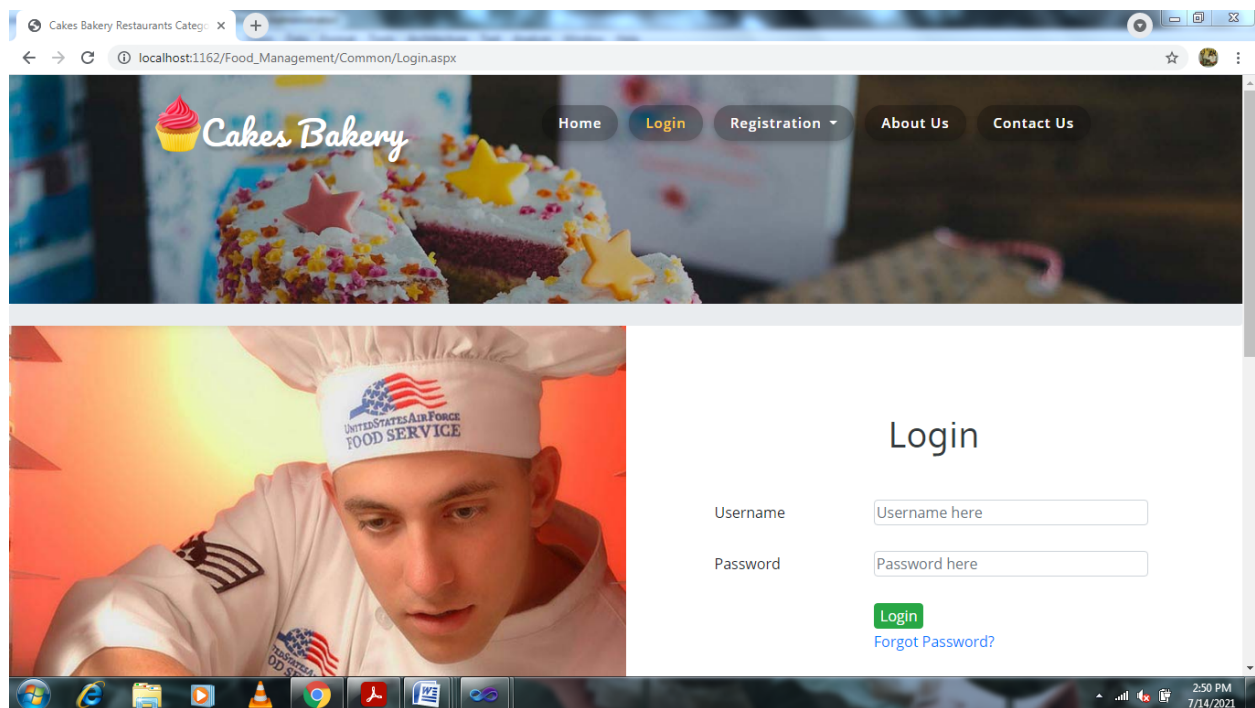
The food management system will help the customers to get good quality and tasty food items from the nearby restaurants at their area. There are also providing food items to deliver at their home / requesting place as per customer’s scheduled time. Customers can also verify the quality of Food / Bakery items on screening the review. Food management system is design to help the customers in finding the nearby restaurants / bakeries.

This system will also ensures the preservation of personal records and ensure the accessibility only by the authorized personal

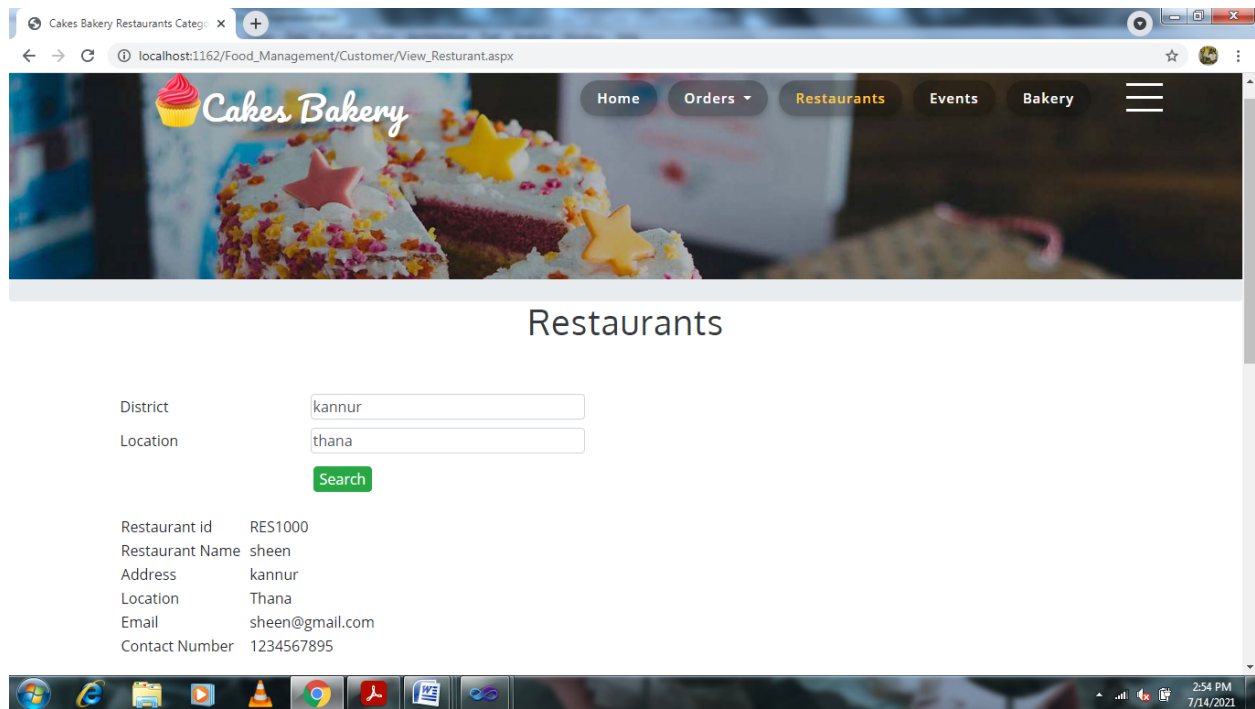
This system has got a better design of scheme because of full feature availability in food management system.

12.

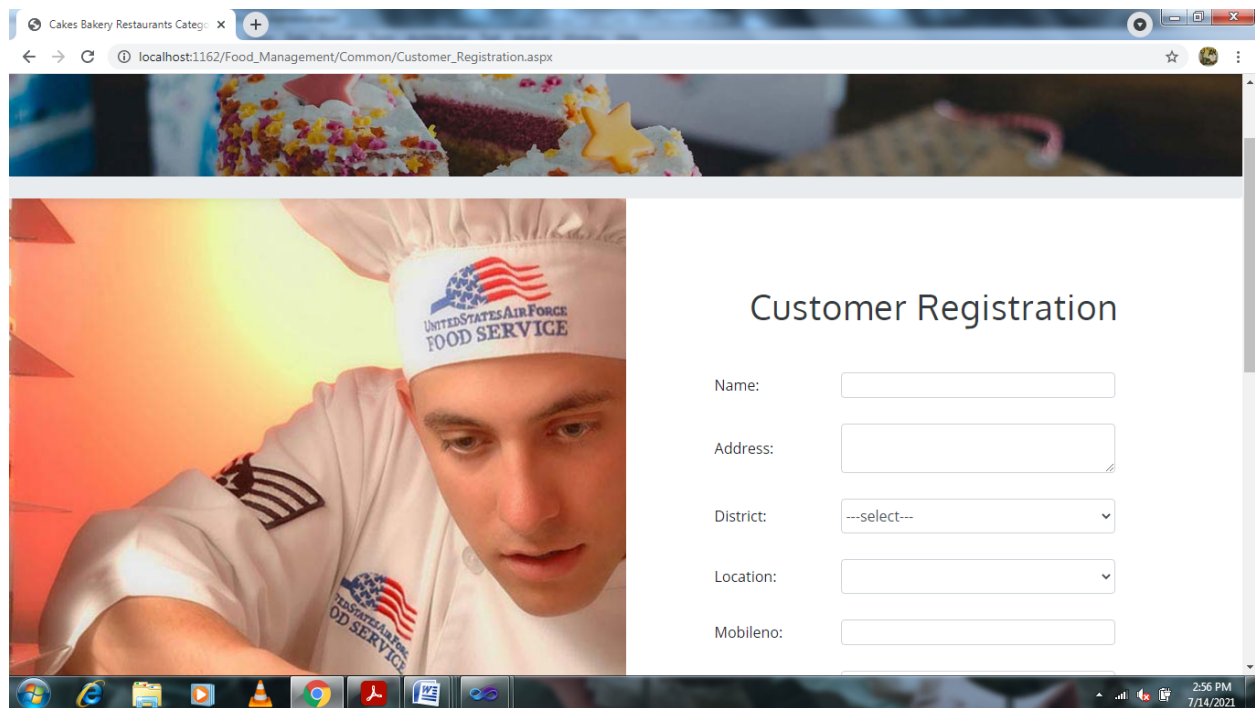
APPENDIX



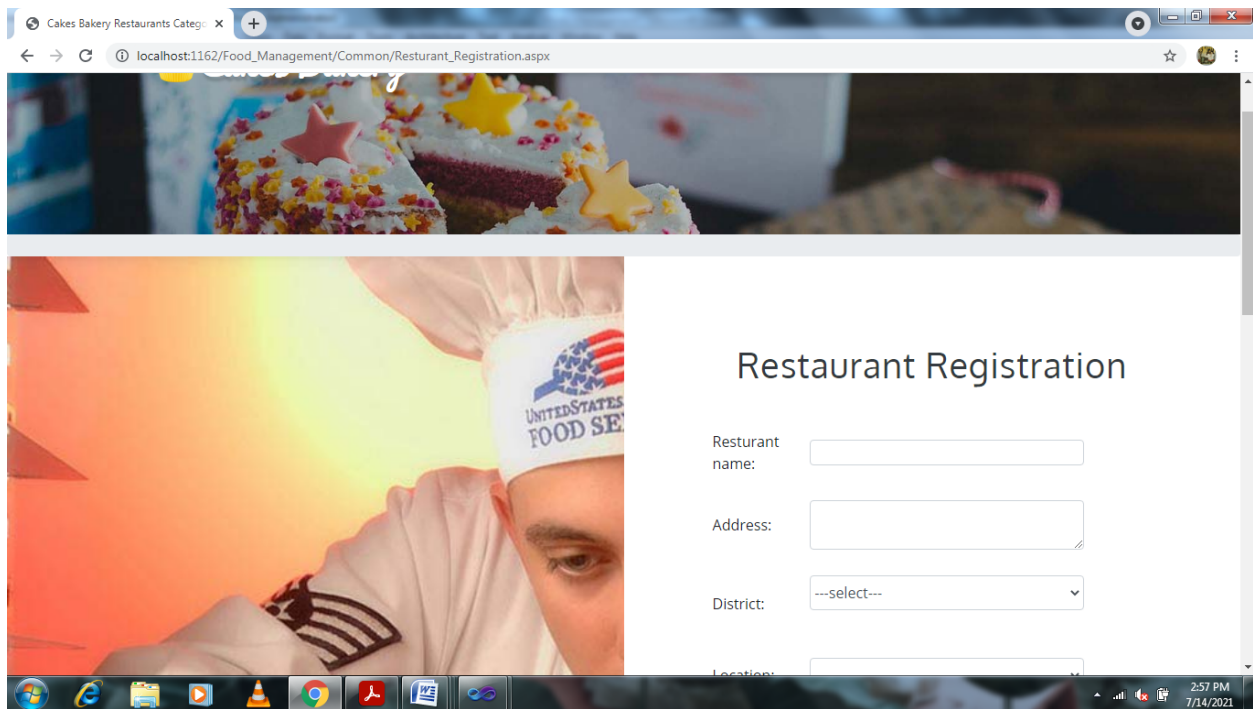
Main Login Page



View All Restaurant list



New Customer Registration



Cakes Bakery Restaurants Catego x +

localhost:1162/Food_Management/Common/Resturant_Registration.aspx

Restaurant Registration

Resturant name:

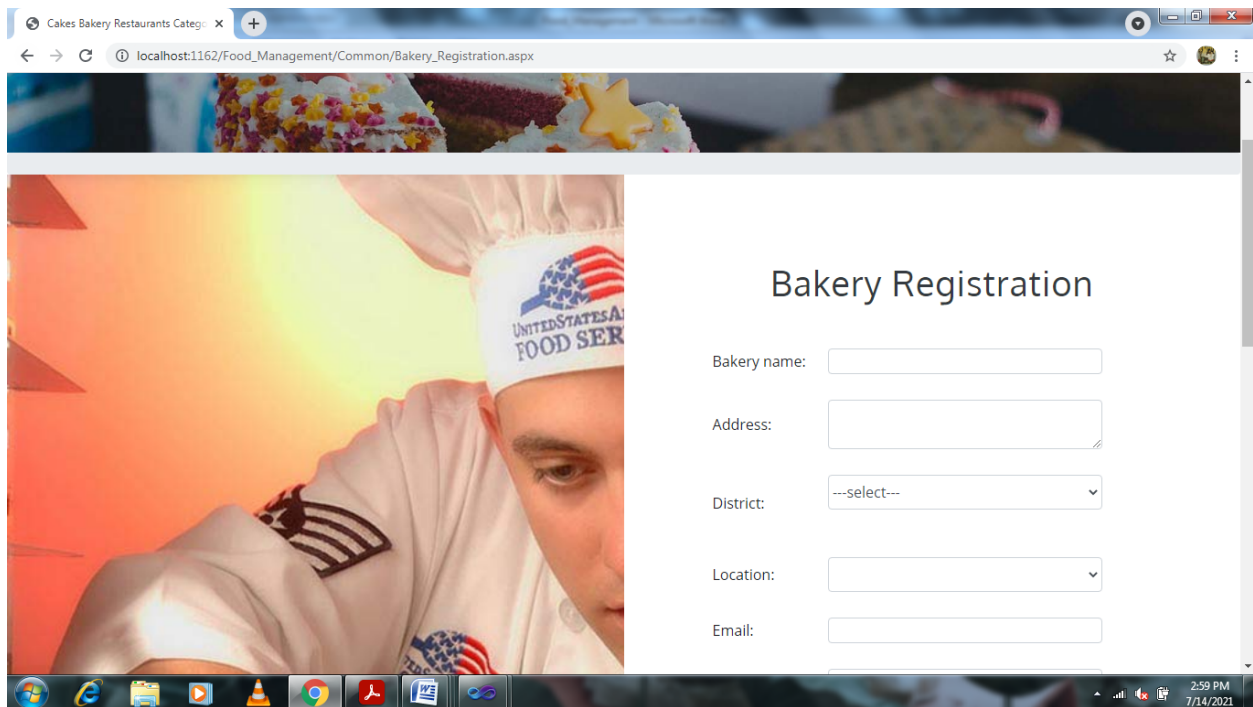
Address:

District: ---select---

Location:

2:57 PM 7/14/2021

New Restaurant Registration



Cakes Bakery Restaurants Catego x +

localhost:1162/Food_Management/Common/Bakery_Registration.aspx

Bakery Registration

Bakery name:

Address:

District: ---select---

Location:

Email:

2:59 PM 7/14/2021

New Bakery Registration

Cakes Bakery Restaurants Catego x +

localhost:1162/Food_Management/Customer/Event_Festival.aspx



Events

Event Title	The Eat Event
Event Description:	make the food dishes seasonal
Location	kannur
Landmark	near S N Park
Starting Date:	15-06-2021
Ending Date:	15-07-2021

Poster



CLASS IX: READING LITERACY

TABLE 1

Musical Instruments to Play

A musical instrument is a device created to make musical sounds. Anything that makes a sound can be used as a musical instrument. The history of musical instruments goes back to the beginning of culture. People first used instruments as music. A human might use a trumpet to signal a war-related event; a drum might be used in a religious ceremony.

Culture also composed and performed a set of sounds called a melody for entertainment. Musical instruments were needed. Some historians report that the earliest musical instrument was a simple flute. Many of the earliest musical instruments were made from animal skins, bones, wood, and other non-metallic materials. Musical instruments were developed separately in the different countries and regions of the world, but when civilizations shared information among themselves, the development of instruments spread. For example, cultures of North America, South America, and Central America used similar instruments and shared their ideas of making instruments that were able to create new.

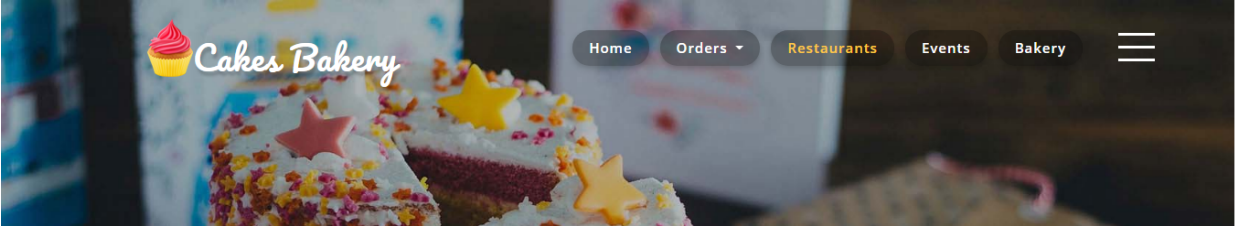
Many different notes have been used to classify instruments over the years. One way to classify instruments is to put them in groups by the range of notes the instruments can play. Another classification is to put them together by what they are made out of. However, the most common method of grouping instruments is to how they produce sounds. The academic study of musical instruments is called ethnomusicology.

3:09 PM
7/14/2021

New / Upcoming Event Festivals

Cakes Bakery Restaurants Catego x +

localhost:1162/Food_Management/Customer/View_My_Order.aspx



Cakes Bakery

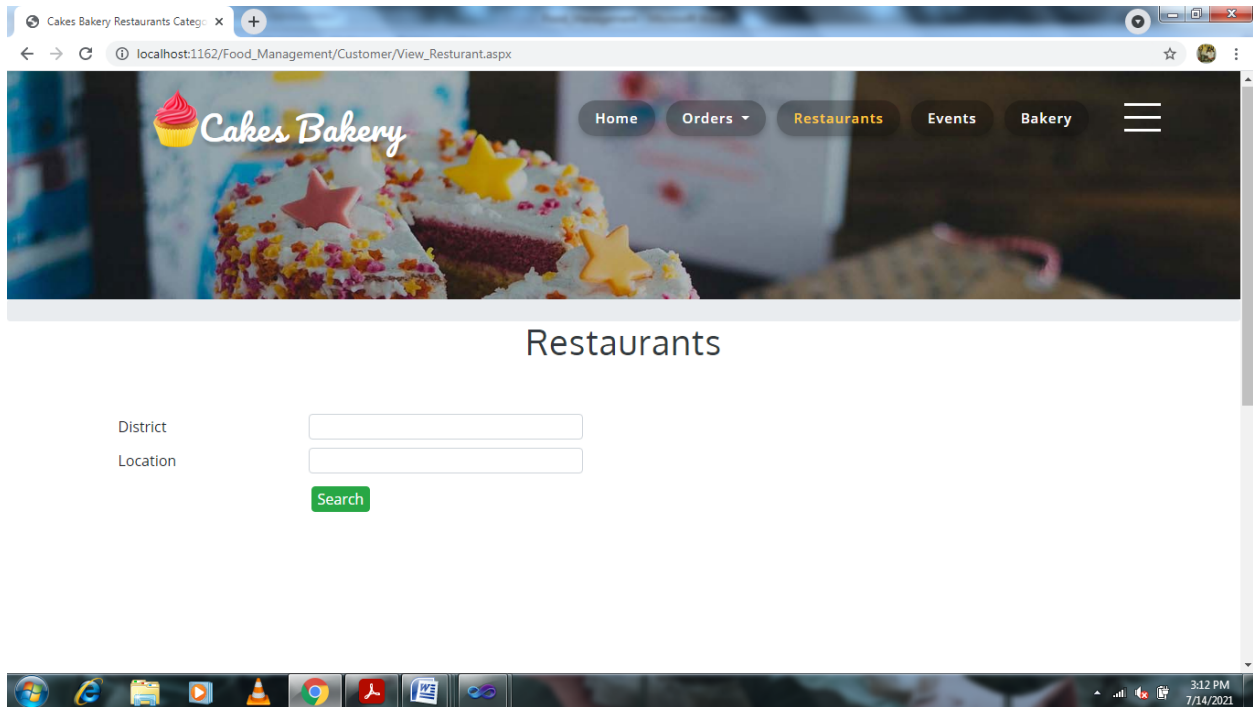
Home Orders Restaurants Events Bakery

Order

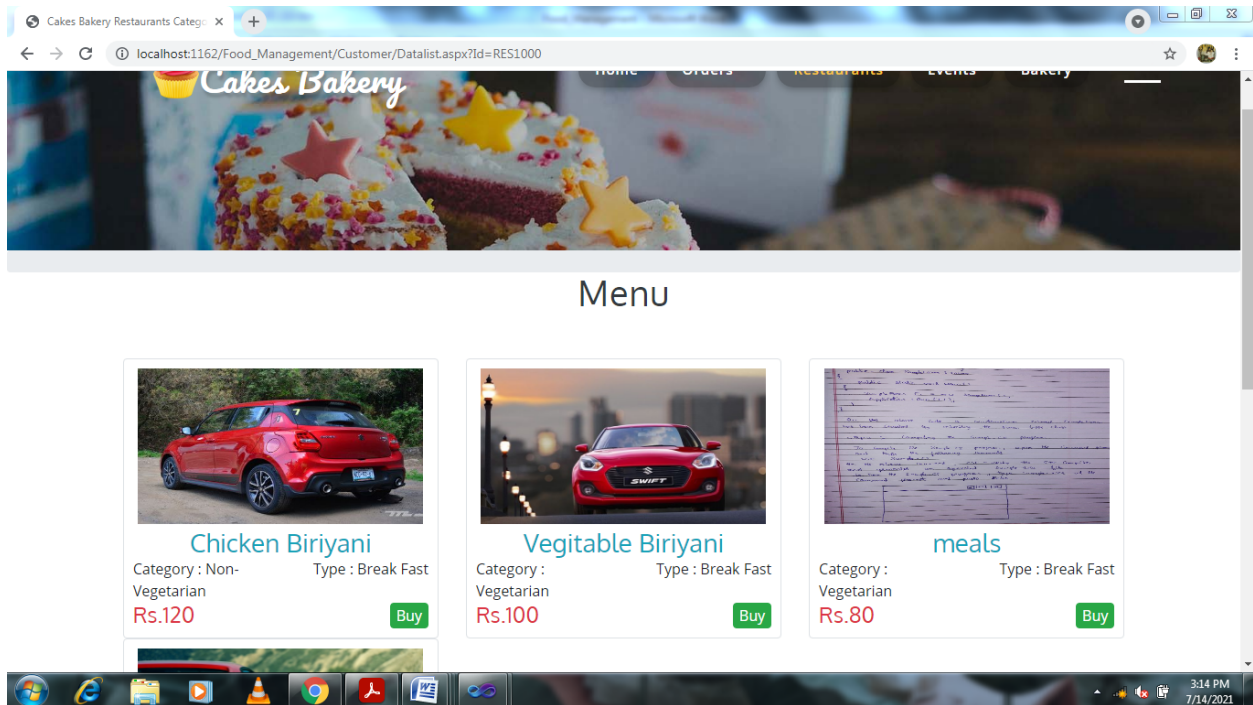
Product Details	
Name	Chicken Biryani
Catagory	Non-Vegetarian
Type	Break Fast
Quantity	2
Rate	240

3:11 PM
7/14/2021

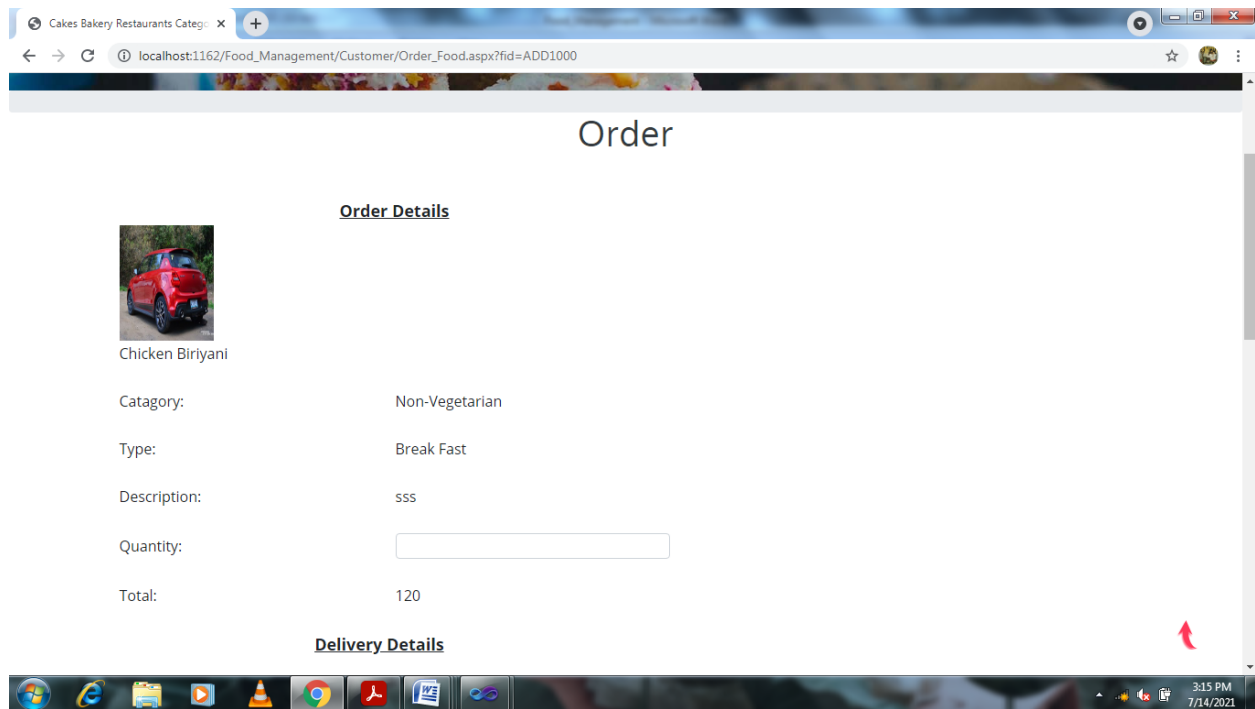
Order Details



Searching for Nearest Restaurants / Bakery



Menu items



Order Details

13.

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